

ATE Survey

Technical Guide

Community attitude, knowledge, and sentiment analysis towards elephants

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Workflow id: **ate_survey**

1. Overview

The **ate_survey** workflow processes Amboseli Trust for Elephants (ATE) community questionnaire responses recorded as **atequestionnaire_rep** events in EarthRanger. It cleans and transforms survey data, computes elephant sentiment scores, runs ANOVA and OLS statistical analyses, and produces a suite of charts, maps, and a Word report.

The workflow delivers:

- 1 demographic summary table (CSV)
- 2 Likert charts — attitudes/beliefs (agree/disagree) and mitigation effectiveness
- 26 pie charts across participant demographic and knowledge questions
- 9 bar charts across tribal, lifestyle, and encounter questions
- 1 elephant sentiment score dataset (CSV) with per-respondent scores
- 1 ANOVA results table (CSV) testing sentiment against gender, age, education, marital status
- 5 boxplots, 2 OLS scatter plots, and 4 Tukey HSD plots
- 3 spatial maps — survey locations, gender distribution, and attitude scores
- 1 Word report (social_survey.docx) populated with all charts and demographic table

Output summary

Output type	Count	Description
Demographic CSV	1	Summary table: gender, age, tribe, household size, education
Likert charts	2	Agree/disagree attitudes and mitigation effectiveness ratings
Pie charts	26	One per survey question (demographics, knowledge, behaviour)
Bar charts	9	Tribal, land tenure, wildlife feelings, encounter frequency
Elephant sentiment CSV	1	Per-respondent overall attitude score
ANOVA results CSV	1	Type II ANOVA: sentiment vs gender, age group, education, marital status
Boxplots	5	Sentiment score distributions by demographic group
OLS scatter plots	2	Sentiment vs age and household size
Tukey HSD plots	4	Post-hoc pairwise comparisons per demographic factor
Spatial maps	3	Survey locations, gender distribution, attitude scores
Word report	1	Populated social_survey.docx with all charts and demographic table

2. Dependencies

2.1 Python packages

Package	Version	Channel
ecoscope-workflows-core	0.22.17.*	ecoscope-workflows
ecoscope-workflows-ext-ecoscope	0.22.17.*	ecoscope-workflows
ecoscope-workflows-ext-custom	0.0.39.*	ecoscope-workflows-custom
ecoscope-workflows-ext-ste	0.0.17.*	ecoscope-workflows-custom
ecoscope-workflows-ext-mnc	0.0.7.*	ecoscope-workflows-custom
ecoscope-workflows-ext-ate	0.0.3.*	ecoscope-workflows-custom

2.2 Connection

Connection	Task	Purpose
EarthRanger	set_er_connection	Fetch atequestionnaire_rep survey event records for the analysis time range

Note: This workflow does not require a Google Earth Engine connection.

2.3 Dropbox files

Four files are downloaded from Dropbox at runtime (**overwrite_existing: false**, retries: 3):

File	Purpose
ate_survey_template_280857.docx	Word report template — populated with charts and demographic table to produce social_survey.docx
Amboseli-Ranch-Boundaries.gpkg	Polygon boundaries for Amboseli ranches — rendered as black outlines on all maps
Amboseli-Swamps.gpkg	Amboseli swamp polygons — rendered in blue (#609cb5) on all maps
National-Parks.gpkg	National park polygons — rendered in green (#4c8c2b) on all maps

2.4 Base maps

Two base map tile layers are configured by default via **set_base_maps_pydeck**:

Layer	URL (abbreviated)	Opacity
ArcGIS World Hillshade	server.arcgisonline.com/.../Elevation/World_Hillshade/MapServer/tile/{z}/{y}/{x}	1.0
ArcGIS World Street Map	server.arcgisonline.com/.../World_Street_Map/MapServer/tile/{z}/{y}/{x}	0.25

2.5 Grouper

The workflow uses an **empty grouper list** (groupers: []). All survey records are processed as a single undivided dataset — no fan-out or per-group branching is performed.

3. Survey Data Pipeline

Survey questionnaire responses are fetched, cleaned, and transformed through an eight-step pipeline before any analysis or charting begins.

3.1 Event retrieval

Parameter	Value
Task	get_events
Event type	atequestionnaire_rep
Columns retained	id, time, event_type, event_category, reported_by, serial_number, geometry, created_at, event_details, patrols
include_details	true
include_null_geometry	false
raise_on_empty	true
include_updates	false
include_related_events	false
include_display_values	false

3.2 Invalid record exclusion

Task: **exclude_value**. Filters out the record with **event_details value 46283** from the dataset. This is a known test/invalid questionnaire entry that must be excluded before processing.

3.3 JSON normalisation

Task: **normalize_json_column**. The **event_details** column contains nested JSON. This step flattens it into individual columns (skip_if_not_exists: true, sort_columns: true), producing columns with the prefix **event_details__**.

3.4 Column mapping

Task: **map_columns**. The normalised DataFrame undergoes significant restructuring:

- **32 columns dropped** — unused fields including event_category, reported_by, end_time, closing_statement, tribe/occupation/land tenure 'if other' free-text columns, and supplementary livestock health detail columns.
- **~60 columns renamed** — all **event_details__** prefixed columns are renamed to human-readable labels (e.g. event_details__participant_age → 'Participant age', event_details__belief_1 → 'Humans and elephants can live together').

3.5 Type conversion

Two conversion tasks run sequentially:

Task	Columns	Count
convert_object_to_value	Participant age, Number of cows, Household size, Land owned (acres), Number of shoats, Agricultural land (acres), Number of sheep or goats, Number of dogs, Number of cattle, Number of donkeys	10
convert_object_to_string	id, all categorical response columns (tribe, gender, knowledge, attitude, belief, exposure, behaviour, mitigation, perception fields)	49

3.6 Missing value fill

Task: **fill_missing_values**. Missing values are filled with type-appropriate defaults:

Column type	Fill value	Count
Numeric (10 columns)	0	Participant age, cow/shoat/livestock counts, land areas, household size
Categorical (49 columns)	Unspecified	All string/response columns

3.7 Survey response mapping

Three **map_survey_responses** tasks normalise raw API response strings to display labels:

Task	Columns	Mapping
map_agree_disagree	11 belief/attitude columns (Humans and elephants can live together, Importance of elephants living here, Elephants should only live inside parks, cost/benefit beliefs, hazard acceptance)	strongly_agree→Strongly agree, agree→Agree, neutral→Neutral, disagree→Disagree, strongly_disagree→Strongly disagree, i_dont_know→I dont know, prefer_not_to_answer→Prefer not to answer
map_true_false	7 elephant knowledge columns (family groups, seasonal movement, head-shaking, trunk-raising, musth behaviour, smell/hearing, female protection)	false→False, true→True, i_dont_know→I dont know
map_no_effect	3 effectiveness columns (crop protection, behaviour practice, water protection)	highly_effective→Highly effective, effective→Effective, not_effective→Not effective, i_dont_know→I dont know

3.8 Ordinal column mapping and integer conversion

map_survey_columns applies custom ordinal ordering to 28 columns (demographic fields plus all Agree/Disagree attitude columns) to ensure correct sort order in charts.

convert_to_int then casts 9 numeric columns to integer type (errors: coerce, fill_value: 0, inplace: false).

4. Demographic Table

Task: **format_demographic_table**. Produces a summary table from the cleaned survey data using the following five columns:

Column
Participant gender
Participant age
Participant tribe
Household size
Highest level of education

The resulting table is persisted to disk as **demographic_table.csv** and is also passed directly to the Word report generator (**demographic_csv: demographic_table.csv**).

5. Column Binning

Task: **bin_columns**. Ten continuous numeric columns are binned into **5 equal-width bins** (suffix: 'bins', inplace: true). The binned columns are used downstream by all chart and map tasks:

Column binned	Resulting column
Participant age	Participant agebins
Number of cows	Number of cowsbins
Household size	Household sizebins
Number of shoats	Number of shoatsbins
Land owned (acres)	Land owned (acres)bins
Agricultural land (acres)	Agricultural land (acres)bins
Number of dogs	Number of dogsbins
Number of cattle	Number of cattlebins
Number of donkeys	Number of donkeysbins
Number of sheep or goats	Number of sheep or goatsbins

Two binned columns are additionally processed by **categorize_bins** to convert the numeric interval labels to ordered categorical dtype:

- Household sizebins
- Participant agebins

6. Charts

6.1 Likert chart — Agree/Disagree

Parameter	Value
Task	create_likert_chart
Input columns (11)	Humans and elephants can live together; Importance of elephants living here; Elephants should only live inside parks; Elephants harm community members; Elephants negatively affect my livelihood; Elephants impact my emotional wellbeing; Benefit from enjoying seeing elephants; Elephants are important for a healthy ecosystem; Acceptable to harm elephants if they damage property or livestock; Acceptable to harm elephants if people are hurt; Receive livelihood benefits from elephants
Title	Agree/Disagree
height_per_question	60 px
min_height	400 px
width	1 200 px
show_percentages	true
sort_questions	false
sort_by	positive
response_order	Strongly disagree → Disagree → Neutral → Agree → Strongly agree
neutral_categories	Neutral
Color: Strongly disagree	#2c5282 (dark blue)
Color: Disagree	#4299e1 (blue)
Color: Neutral / Unspecified / I dont know	#a0aec0 (grey)
Color: Agree	#ed8936 (orange)
Color: Strongly agree	#c05621 (dark orange)

Output: **elephants_relationship_likert.html**

6.2 Likert chart — Mitigation Effectiveness

Parameter	Value
Input columns (3)	Effectiveness rating of primary crop protection method; Effectiveness rating of that practice; Effectiveness rating of water protection
Title	Effective/Not effective

Parameter	Value
response_order	Not effective → Effective → Highly effective
neutral_categories	I dont know
Color: Not effective	#D64545 (red)
Color: I dont know	#A8A8A8 (grey)
Color: Unspecified	#D4D4D4 (light grey)
Color: Effective	#2A9D8F (teal)
Color: Highly effective	#2E7D32 (dark green)

Output: **effectiveness_mitigation_methods.html**

6.3 Pie charts

Task: **draw_pie_and_persist**. Generates one pie chart per column (26 total). Charts are saved as individual HTML files then converted to PNG (25 ms timeout, scale 2x). Columns:

Column
Participant gender
Female elephants live in family groups
Female elephants protect their young
Participant age group
See more elephants now than before
Do you change routes or schedules because of elephants
What do you do when you encounter elephants on foot
Noticed signs of illness in wild animals
Willingness to join future community dialogue
Use measures to protect crops from elephants
Ever been involved in or witnessed an elephant harmed
Elephants can smell and hear from far away
Elephants move seasonally for food and water
Elephant may shake its head when annoyed
Elephant signals awareness by raising trunk
Male elephants with secretions may be more aggressive
Highest level of education
Use measures to protect livestock from elephants

Column
Protect water sources from elephants
Marital status
Greatest threat to your livestock
Benefit from the KCCDT Big Life electric fence
Pay into fence maintenance fund
Do you report elephant conflict incidents
Which intervention would help you in future
Opinion on having elephants in the area

6.4 Bar charts

Task: **draw_bar_and_persist**. Generates one bar chart per column (9 total). Saved and converted to PNG (25 ms timeout, scale 2x). Columns:

Column
Participant tribe
Years living in area
Overall feelings about wildlife
How often seen elephants in the last year
Reaction to hearing about elephants being harmed
Land tenure
Greatest threat to crop production
Household sizebins
Participant agebins

7. Elephant Sentiment Score

A per-respondent **elephant_sentiment_score** is computed by **calculate_elephant_sentiment_score** from 11 attitude/belief columns. Positive and negative columns are weighted in opposite directions to produce an overall attitude score.

7.1 Input columns

Direction	Column
Positive	Humans and elephants can live together
Positive	Importance of elephants living here
Positive	Benefit from enjoying seeing elephants
Positive	Elephants are important for a healthy ecosystem
Positive	Receive livelihood benefits from elephants
Negative	Elephants should only live inside parks
Negative	Elephants harm community members
Negative	Elephants negatively affect my livelihood
Negative	Elephants impact my emotional wellbeing
Negative	Acceptable to harm elephants if they damage property or livestock
Negative	Acceptable to harm elephants if people are hurt

Prior to the score calculation, geometry outliers are removed (**z_threshold: 3**) and the 11 attitude columns plus id and geometry are filtered to form the input GeoDataFrame. The resulting scores are persisted to disk as **elephant_sentiment_scores.csv**.

8. Statistical Analysis

The workflow runs three complementary statistical analyses to investigate which demographic factors predict elephant sentiment scores.

8.1 ANOVA

Task: **perform_anova_analysis**. A **Type II ANOVA** is run with **elephant_sentiment_score** as the target and four demographic factors as predictors:

Factor column
gender_of_participant
age_group_distribution
education_level
marital_status

Results are persisted as **anova_results.csv**.

8.2 Boxplots

Task: **draw_boxplot_and_persist**. One boxplot per demographic column (5 total), with **elephant_sentiment_score** on the y-axis:

Column (x-axis)
marital_status
education_level
tribe
age_group_distribution
gender_of_participant

8.3 OLS scatter plots

Task: **draw_ols_scatterplot_and_persist**. Ordinary least-squares regression plots with **elephant_sentiment_score** on the y-axis for 2 continuous predictors:

Column (x-axis)
age
household_size

8.4 Tukey HSD plots

Task: **draw_tukey_plots_and_persist**. Post-hoc pairwise comparison plots for 4 factor columns against **elephant_sentiment_score**:

Column
education_level
age_group_distribution
gender_of_participant
marital_status

All statistical charts are saved as individual HTML files then converted to PNG (25 ms timeout, device_scale_factor: 2.0, max_concurrent_pages: 1).

8.5 Merged stats DataFrame

Before statistical analysis, a merged DataFrame is assembled: demographic columns (id, Participant age, Marital status, Highest level of education, Participant tribe, Household size, Participant age group, Participant gender) are joined to the sentiment scores on **id** (left join, preserve_left_index: false). Columns are then renamed to snake_case for use in statistical tasks.

9. Spatial Maps

Three EcoMaps are produced. All share the same base map tiles, static context layers, and map viewport. Each uses a scatter plot layer with radius 4 m, opacity 0.75, stroked: true, and line_color [0,0,0].

9.1 Static context layers

Three polygon GeoPackage files are downloaded, loaded, reprojected to EPSG:4326, and typed via `get_gdf_geom_type`. All three are included on every map via `combine_deckgl_map_layers`:

Layer	Fill / line color	Hex	Line width	Opacity
Ranch boundaries	[0, 0, 0]	#000000	2.25	0.45 (outline only, filled: false)
Amboseli swamps	[96, 156, 181]	#609cb5	0.5	0.45 (filled: true)
National parks	[76, 140, 43]	#4c8c2b	0.75	0.45 (filled: true)

Note: The map viewport is zoomed to the extent of the reprojected ranch boundaries GeoDataFrame (pitch: 0, bearing: 0). All maps use max_zoom: 10 and legend placement: bottom-right.

9.2 Survey locations map

Parameter	Value
Input data	bin_survey_cols GeoDataFrame (z_threshold: 3 outlier removal)
Fill color	[199, 0, 57] — crimson red (#C70039), single colour for all points
Legend	Survey location (#C70039)
Output	survey_locations_ecomap.html / .png
PNG timeout	40 000 ms

9.3 Gender distribution map

Parameter	Value
Input data	bin_survey_cols GeoDataFrame (z_threshold: 3 outlier removal)
Color column	Participant gender → gender_colors (apply_color_map)
Colormap (3 values)	#FFC0CB (pink), #0000FF (blue), #DC143C (crimson)
Legend title	Gender
Output	gender_distribution_ecomap.html / .png
PNG timeout	40 000 ms

9.4 Attitude scores map

Parameter	Value
Input data	calc_attitude_scores GeoDataFrame (z_threshold: 3 outlier removal)
Color column	overall_attitude → attitude_colors (apply_color_map)
Colormap (5 values)	#FF0000 (red), #FFA500 (orange), #FFFF00 (yellow), #00FF00 (green), #0000FF (blue)
Legend title	Attitude scores
Legend columns	label_column: overall_attitude, color_column: attitude_colors, sort: ascending
Output	attitude_scores_community.html / .png
PNG timeout	40 000 ms

10. Word Report

Task: **persist_survey_word**. The downloaded Word template (**ate_survey_template_280857.docx**) is populated with survey outputs and saved as **social_survey.docx**.

Parameter	Value
template_path	ate_survey_template_280857.docx (downloaded from Dropbox)
output_dir	ECOSCOPE_WORKFLOWS_RESULTS
time_period	Analysis time range (from set_time_range)
box_h_cm	6.5 cm — height of image placeholder boxes
box_w_cm	11.11 cm — width of image placeholder boxes
filename	social_survey.docx
demographic_csv	demographic_table.csv

Note: The Word template contains placeholder boxes sized to box_h_cm × box_w_cm. The task inserts charts and the demographic table into these placeholders.

11. Output Files

All outputs are written to **ECOSCOPE_WORKFLOWS_RESULTS**. Files marked with *<column>* are produced once per chart column.

File	Description
demographic_table.csv	Summary demographic table (gender, age, tribe, household size, education)
elephant_sentiment_scores.csv	Per-respondent overall_attitude score computed from 11 belief/attitude columns
anova_results.csv	Type II ANOVA results: elephant_sentiment_score vs 4 demographic factors
elephants_relationship_likert.html / .png	Agree/Disagree Likert chart for 11 attitude columns
effectiveness_mitigation_methods.html / .png	Effective/Not effective Likert chart for 3 mitigation effectiveness columns
_pie.html / .png (×26)	One pie chart per survey question
_bar.html / .png (×9)	One bar chart per survey question
_boxplot.html / .png (×5)	Sentiment score boxplot per demographic group
_ols.html / .png (×2)	OLS regression scatter plot (age, household_size vs sentiment)
_tukey.html / .png (×4)	Tukey HSD post-hoc comparison plot per demographic factor
survey_locations_ecomap.html / .png	All survey point locations on Amboseli base map
gender_distribution_ecomap.html / .png	Survey points coloured by respondent gender
attitude_scores_community.html / .png	Survey points coloured by overall elephant attitude score
social_survey.docx	Final Word report populated with all charts and demographic table

12. Workflow Execution Logic

12.1 Global skip conditions

All tasks carry the default skipif block (**task-instance-defaults**):

Condition	Behaviour
any_is_empty_df	Skip task if any input DataFrame is empty
any_dependency_skipped	Skip task if any upstream dependency was skipped

12.2 No mapvalues fan-out

This workflow processes all survey records as a single batch. There is no **mapvalues** or **split_groups** fan-out — every task runs exactly once on the full dataset. The empty grouper list (groupers: []) is passed through to the dashboard but is not used for any data splitting.

12.3 Screenshot timing

Chart / map type	wait_for_timeout	Reason
Pie charts (26)	25 ms	Static Plotly HTML — no tiles
Bar charts (9)	25 ms	Static Plotly HTML — no tiles
Boxplots (5)	25 ms	Static Plotly HTML — no tiles
OLS scatter plots (2)	25 ms	Static Plotly HTML — no tiles
Tukey plots (4)	25 ms	Static Plotly HTML — no tiles
Likert charts (2)	25 ms	Static Plotly HTML — no tiles
Attitude scores map	40 000 ms	Tile map — waits for ArcGIS tile rendering
Gender distribution map	40 000 ms	Tile map — waits for ArcGIS tile rendering
Survey locations map	40 000 ms	Tile map — waits for ArcGIS tile rendering

Note: All PNG conversions use `device_scale_factor`: 2.0 and `max_concurrent_pages`: 1. The two Likert HTMLs are passed as a list to a single `html_to_png` call.

12.4 Dashboard

The workflow concludes with **gather_dashboard** which packages workflow details, time range, and groupers. The **widgets** list is empty — no single-value or map widgets are configured for this workflow.

13. Software Versions

Package	Version pinned in spec.yaml
ecoscope-workflows-core	0.22.17.*
ecoscope-workflows-ext-ecoscope	0.22.17.*
ecoscope-workflows-ext-custom	0.0.39.*
ecoscope-workflows-ext-ste	0.0.17.*
ecoscope-workflows-ext-mnc	0.0.7.*
ecoscope-workflows-ext-ate	0.0.3.*